

Examining Network Switches

At the end of this episode, I will be able to:

1. Identify and explain the functionality and purpose of network switches.

Learner Objective: *Identify and explain the functionality and purpose of network switches.*

Description: In this episode, the learner will

- Introduction
 - o Definition: A network switch is a network device, connecting devices inside local area networks (LAN) and uses MAC addresses to forward data to its destination.
- Operational Principles
 - o Frame Forwarding: How switches use MAC addresses to determine the destination of a frame.
 - o Collision Domains: A network segment where only one device can transmit at a time. Switches reduce collisions and improving network performance.
- Types of Switches
 - o Unmanaged Switches: Plug and play switches without configuration.
 - o Managed Switches: Offer features for configuration, VLAN setup, monitoring, etc.
 - o Layer 2 (Data Link) Switches: Operate using MAC addresses.
 - o Layer 3 (Network) Switches: Incorporate routing functions and operate using both MAC and IP addresses.
 - o Multilayer Switches: Operate at multiple layers, often combining Layer 2 and Layer 3 functions.
 - o PoE (Power over Ethernet) Switches: Deliver power and data over a single Ethernet cable, useful for devices like IP phones and cameras.
- Benefits of Using Switches
 - o Efficiency: Reduced collisions and dedicated bandwidth for each connection.
 - o Performance: High-speed connectivity, especially in full-duplex mode.
 - o Flexibility: Scalability to accommodate growing networks.
 - o Security: Features like VLANs can segment network traffic.
- Store-and-Forward Switching
 - o Process - In store-and-forward switching, the switch receives the entire data frame into its buffer. It checks it for errors (usually using a Cyclic Redundancy Check, or CRC) before forwarding it.
 - o Error Checking - This method allows the switch to catch and discard frames with errors, reducing the amount of erroneous data transmitted through the network.
 - o Latency - It introduces more latency compared to other methods, as the switch must wait to receive the entire frame before beginning to forward it.
 - o Advantages - High data integrity since frames are checked for errors.
 - o Disadvantages - Higher latency, especially with larger frames.
- Cut-Through Switching
 - o Process - Cut-through switching begins forwarding the frame immediately, as the switch reads the destination MAC address without waiting to receive the entire frame.
 - o Error Checking - There is minimal error checking with this method. The switch does not check the CRC so that it may forward corrupted frames.
 - o Latency - This method has lower latency compared to store-and-forward since the switch starts forwarding the frame almost immediately.
 - o Advantages - Lower latency, making it suitable for time-sensitive applications.
 - o Disadvantages - Potential for forwarding corrupted frames, as there is no error checking.
- Fragment-Free Switching
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- Process - Fragment-free switching is a variation of cut-through switching. The switch waits for the first 64 bytes of the frame (the collision window) before forwarding.
 - o Error Checking - By waiting for the first 64 bytes, the switch can avoid forwarding most collision fragments (runt). However, it still doesn't check the entire frame.
 - o Latency - It has slightly higher latency than pure cut-through but less than store-and-forward.
 - o Advantages - Balances the benefits of cut-through (low latency) and store-and-forward (error checking), reducing the chances of forwarding error frames.
 - o Disadvantages - There is still a possibility of forwarding corrupted frames, although it's lower than in cut-through switching.
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Ethernet Frame

- Preamble
 - o This section alerts the receiving system of the incoming frame's start and facilitates synchronization.
 - SFD (Start Frame Delimiter)
 - o Indicates that the Destination MAC Address field commences with the subsequent byte.
 - Destination MAC
 - o Specifies the receiver of the frame.
 - Source MAC
 - o Specifies the originator of the frame.
 - Type
 - o Designates the protocol type encapsulated within the frame, such as IPv4 or IPv6.
 - Data and Pad
 - o Holds the frame's payload. Padding is used to fulfill the minimum length requirement of this field, which is 46 bytes.
 - FCS (Frame Check Sequence)
 - o Includes a 32-bit Cyclic Redundancy Check (CRC), used for identifying any data corruption.
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- Additional Resources:
 - o If applicable